HW02

BFS1 (p.1)

Root depth 0; exclude repeated states. Initial board: $\begin{bmatrix} 2 & 8 & 3 \\ 1 & 6 & 4 \\ 7 & \square & 5 \end{bmatrix}$

Depth d	0	1	2	3	4	5
States Nd	1	3	5	10	14	28

Blank degree: corner2, edge3, center4. Exclude parent reversals.

Depth	Corner blanks	Edge blanks	Centerblanks	Total
0	0	1	0	1
1	2	0	1	3
2	0	5	0	5
3	8	0	2	10
4	0	14	0	14
5	20	0	8	28

$N_2 = 2(2-1)+(4-1)=5$, $N_3=5(3-1)=10$.

Half of depth 5 is examined:

$$N=1+3+5+10+14+\frac{28}{2}=47 \text{ states.}$$

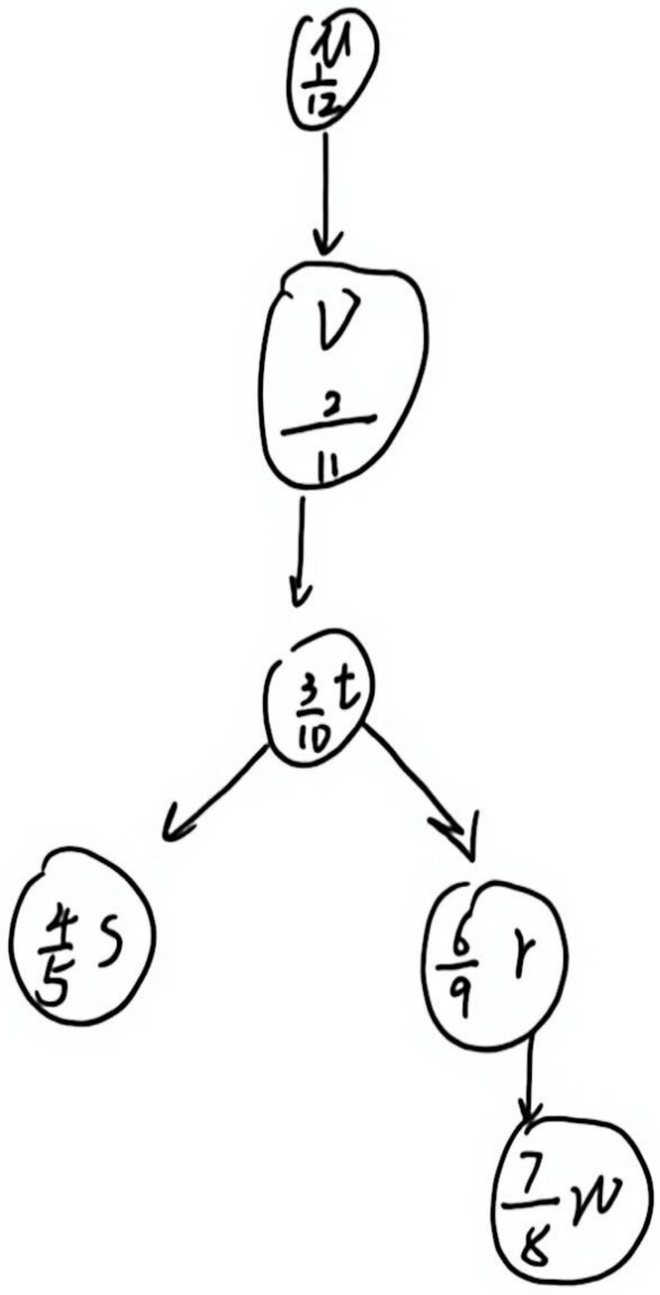
Root and goal included.

DFS1(p2)

Discovery order:

$u \to v \to t \to s$, return to t, $t \to r \to w$,

The tree edges are (u,v), (v,t), (t,s), (t,r), (r,w).



Labels: discovery/finish. times; double circle: backtrack node.

Returns:

$s \to t$, $w \to r \to t \to v \to u$.

Backtrack node = return point starting a new unvisited branch.

bt=t

$E_{back} = \{(s,u), (w,v)\}$; all remaining edges are tree edges.

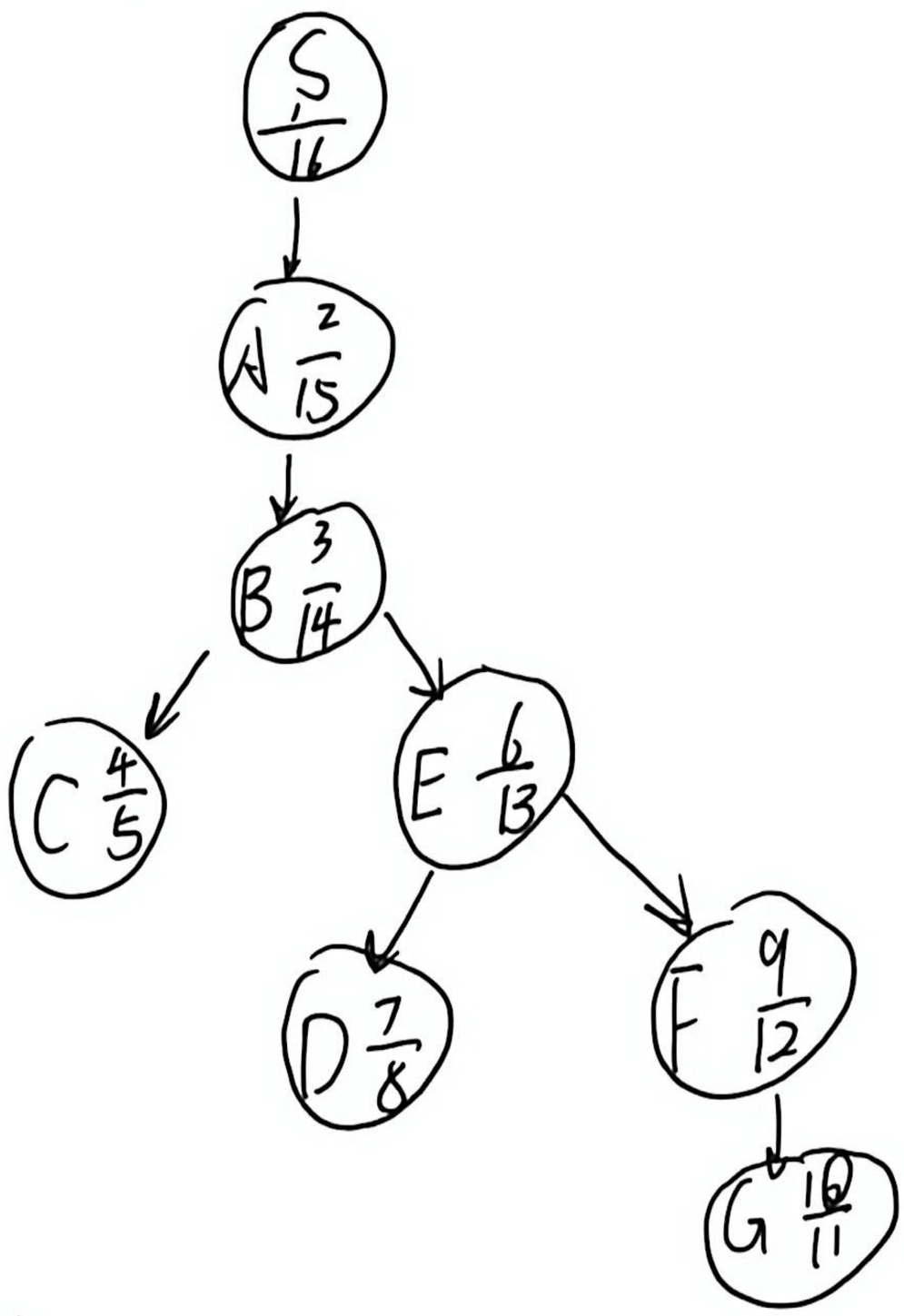
DFS 2 (p.3)

start S; alphabetical adjacency; right-hand graph. D is one vertex, visited once.

Discovery order:

S, A, B, C, E, D, F, G

Labels: discovery/finish times; double circles: backtrack nodes.



bt = {B, E}

C ⇝ B → E, D → E → F

D ∈ visited. when A, S resume: no second D child.

DFS3 (p.4)

Keep the supplied prefix. Convention for remaining neighbors: ascending rc labels (rc = 10r+c). The grid slide gives no fixed directional order.

1. Backtrack to $bf_1 = 41$:

$$(3,2) \to (2,2) \to (1,2) \to (1,1) \to (2,1) \to (3,1) \to (4,1)$$

Visit:

$$(4,1) \to (5,1) \to (6,1) \to (6,2)$$

62 dead end (all neighbors visited).

2. Backtrack to $bf_2 = 42$:

$$(6,2) \to (6,1) \to (5,1) \to (4,1) \to (4,2)$$

Visit:

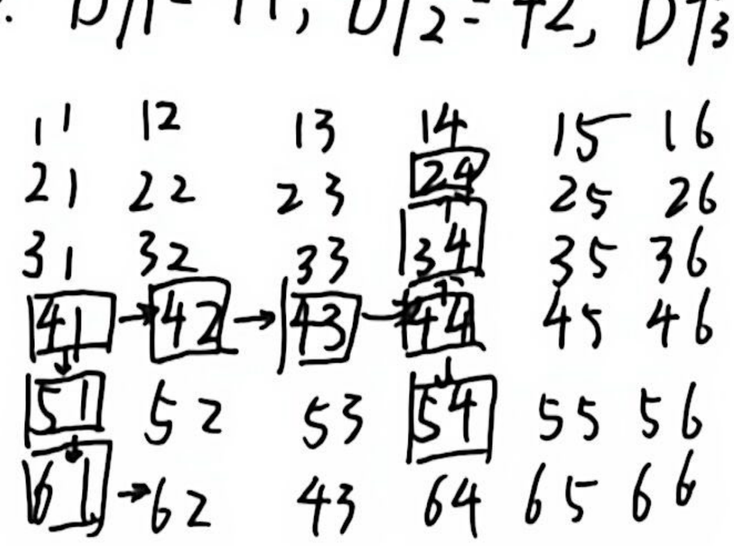
$$(4,2) \to (4,3) \to (4,4) \to (3,4) \to (2,4)$$

$34 < 54$, so visit 34 first; 24 is a dead end.

3. Backtrack to $bf_3 = 44$:

$$(2,4) \to (3,4) \to \boxed{bf_3 = (4,4)}$$

$44 \to 54$: goal; stop. $bf_1 = 41$, $bf_2 = 42$, $bf_3 = 44$



Arrows: new visit.

$$33 \to 23 \to 13 \to 14 \to 15 \to 25 \to 35 \to 45 \to 55$$
$$\to 65 \to 64 \to 63 \to 53 \to 52 \to 42 \to 43 \to 44 \to 54$$

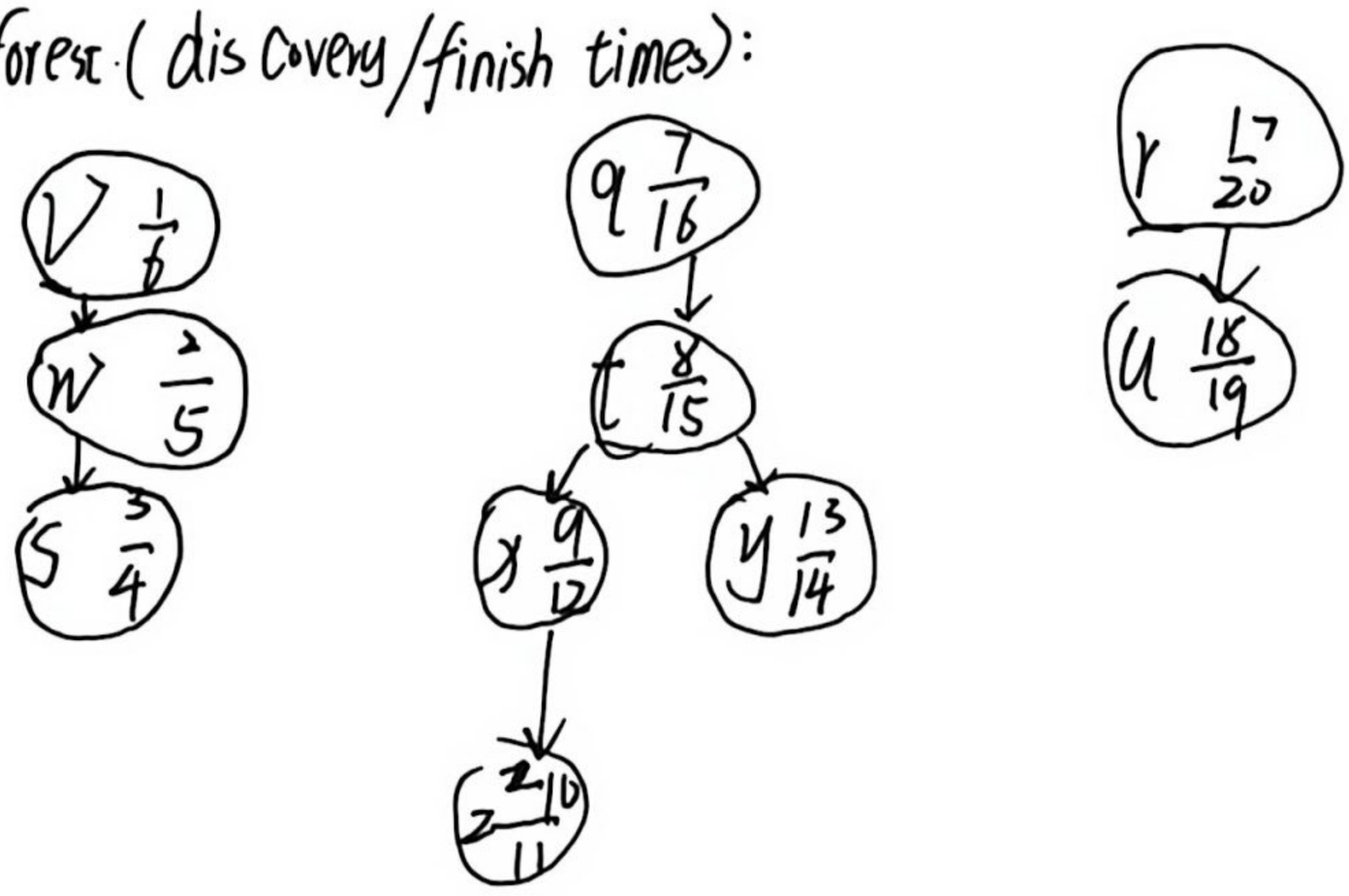
DFS4 (p.5)

start v: remaining roots and adjacency lists alphabetical

vertex	Outgoing neighbors	Vertex	Outgoing neighbors
q	s, t, w	r	u y
s	v	t	x y
u	y	v	w
w	s	x	z
y	q	z	x

Discovery order: v, w, s q, t, x, z, y, r, u

DFS forest (discovery/finish times):



Finish order: s, w, v, z, x, y, t, q, u, r. The nontree edges are:

Back edges: s→v, z→x, y→q.

Cross edges: q→s, q→w, u→y, r→y.

Roots: v, q, r

DFSS (p, 6)

Reverse DFs finish order:

p p, o, s, m, r, y, v, x, w, z, u, q, t.

Check: $(u,v) \in E \Rightarrow pos(u) < pos(v)$.

Vertex	outgoing neighbors (in the ordering later)
p	o s z
n	o q u
o	r s v
s	r
m	q r x
r	u y
y	v
v	w x
w	z
u	t
q	t'

t, x, z : no outgoing edges

Dijkstral (p.7)

Source a; w(b,h) = 1. Initially d(a)=0, all other d = ∞

d(v) ← min{d(v), d(u)+w(u,v)}; settle minimum d (alphabetical ties).

step	settled	a	b	h	g	f	c	i	e	d.
0	–	0	∞	∞	∞	∞	∞	∞	∞	∞
1	a	0	4	8	∞	∞	∞	∞	∞	∞
2	b	0	4	5	∞	∞	12	∞	∞	∞
3	h	0	4	5	6	∞	12	12	∞	∞
4	g	0	4	5	6	8	12	12	∞	∞
5	f	0	4	5	6	8	12	12	18	22
6	c	0	4	5	6	8	12	12	18	19
7	i	0	4	5	6	8	12	12	18	19
8	e	0	4	5	6	8	12	12	18	19
9	d	0	4	5	6	8	12	12	18	19

key relaxations:

d(g) = d(h) + 1 = 6

d(f) = d(g) + 2 = 8

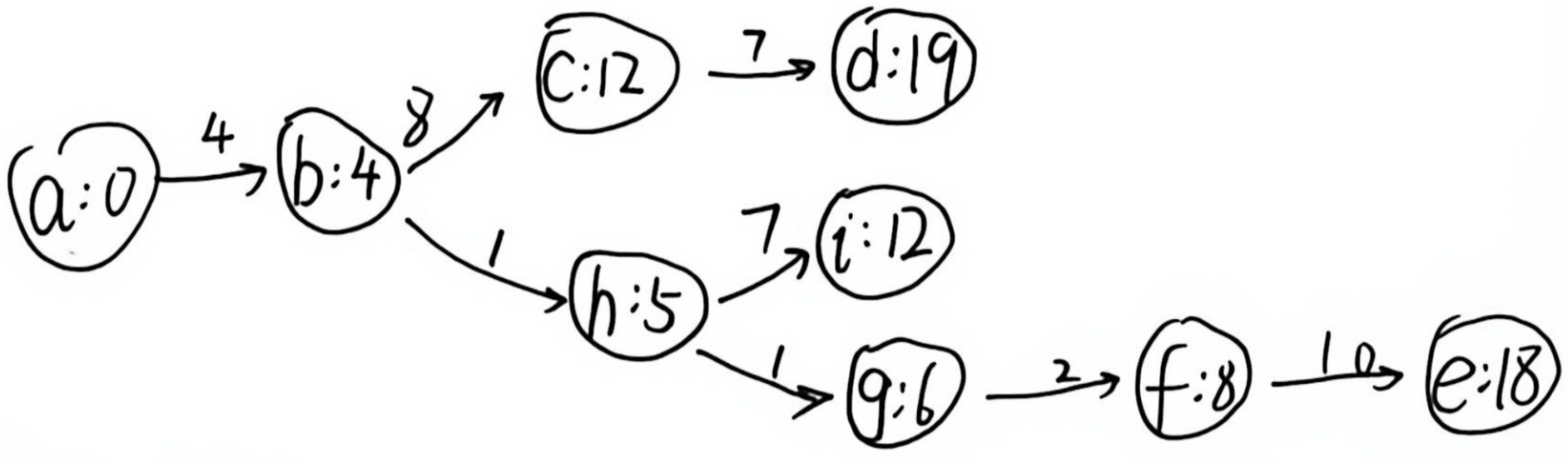
d(d) = d(f) + 14 = 22 initially,

d(d) = min(22, d(e) + 7) = min(22, 19) = 19.

d(i) = d(h) + 7 = 12

d(e) = d(f) + 10 = 18

Final shortest-path tree:



Node labels: vertex and distance.

A star I (p.8)

$f(n) = g(n) + h(n)$; g = best known cost from A.

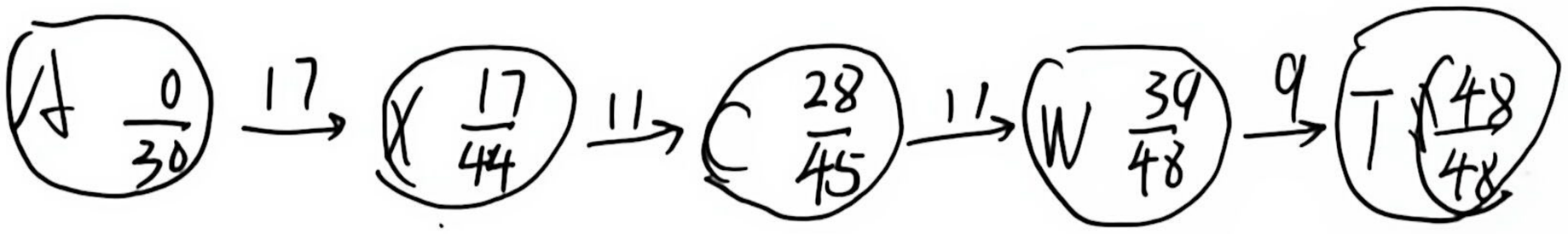
n	A	X	B	C	L	W	K	T	S	H
h(n)	30	27	16	17	33	9	19	0	17	22

OPEN = {A(0,30)}. Notation n(g,f); OPEN sorted by f.

step	Pop	Calculations	OPEN after step
1	A	g(X) = 0+17 = 17, f(X) = 17+27 = 44	X (17,44)
2	X	g(B) = 17+16 = 33, f(B) = 49;	C (28,45), B (33,49)
		g(C) = 17+11 = 28, f(C) = 45.	
3	C	g(W) = 28+11 = 39, f(W) = 48	W (39,48), B (33,49)
		g(L) = 28+20 = 48, f(L) = 81.	L (48,81)
4	W	g(T) = 39+9 = 48, f(T) = 48; g(B) = 51 > 33 (rej	T (48,48), B (33,49)
		g(K) = 39+10 = 49, f(K) = 68	K (49,68), L (48,81)
5	T	Goal. popped; stop	B (33,49), K (49,68) L (48,81)

CLOSED: A, X, C, W, T.

A → X → C → W → T, 17+11+11+9 = 48



Node labels: g/f. Stop when the goal is popped

↓ star2 (p.9)

h(B) = 6; relax and update OPEN entries and parents.

Step	Pop	Calculations	OPEN after step
1	A	g(X)=17, f(X)=44	X(17,44
2	X	g(B)=33, f(B)=33+6=39, g(C)=28, f(C)=45	B(33,39), C(28,45)
3	B	g(W)=33+12=45, f(W)=54 g(T)=33+17=50 f(T)=50 g(S)=33+7=40, f(S)=57	W(45,54), S(40,57)
4	C	g(W)=45→39, f(W)=48, p(W)=C g(L)=48, f(L)=81	W(39,48), T(50,50) S(40,57), L(48,81)
5	W	g(T):50→48, p(T)=W, g(K)=49 f(K)=68; g(B via L)=>33 (rej)	T(48,48), S(40,57) K(49,68), L(48,81)
6	T	Goal popped; stop	S(40,57), K(49,68), L(48,81)

Result:

A → X → C → W → T with cost 48.

Expansion order: A, X, B, C, W, T.

Without OPEN relaxation: W(45,54) stays stale; T(50,50) is popped, giving the wrong path A → X → B → T (cost 50>48)

$h(B) \in \{6, 16\} \leq 17 = h^*(B)$. admissible. With h(B)=6, consistency fails.

$h(X) = 27 > 16 + h(B) = 22$, $h(S) = 17 > 7 + h(B) = 13$.

Inconsistent heuristic: reopen CLOSED if a better g appears.

Astar3 (p.10)

(a) Continuous positions: $S \subset \mathbb{R}^2$, $|S| = \infty$

Visibility graph: Vertices S, G and obstacle corners;

edges are unobstructed straight segments

$5+3+4+4+4+3+6 = 29$ vertices

$|\text{Visibility}| = 29+2 = 31$

(b) Triangle inequality: $|AC| \leq |AB| + |BC|$

Straighten free-space bends; essential bends occur at obstacle vertices.

Shortest path: straight segments through visible corners;

some segments follow boundary edges.



Illustration of the geometric principle, not the original scene

Assume a point robot, Euclidean distance, boundary contact allowed.